

External Shock Propagation to Hong Kong Under the Currency Board: A VAR Analysis of US Monetary Policy and China Growth Spillovers

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Abstract

How do US monetary policy shocks and China growth shocks propagate to Hong Kong's real economy under the currency board? I estimate a six-variable quarterly VAR for Hong Kong (1998Q1–2024Q3) with an external-first Cholesky identification scheme. Forecast error variance decomposition reveals a sharp division of influence: China GDP growth explains 60–78% of Hong Kong GDP forecast error variance at the two-year horizon, while the US federal funds rate accounts for less than 5%. The pattern reverses for interest rates, where the US FFR explains 59–75% of HIBOR variance through the currency board peg. Historical decomposition attributes the 2008–09 recession primarily to China growth shocks and the post-2022 HIBOR spike to US monetary tightening. Results are robust to alternative Cholesky orderings and Bayesian VAR shrinkage.

Keywords: Hong Kong, currency board, VAR, FEVD, monetary transmission, China spillovers

JEL Classification: E32, E52, F41, F42

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1 Introduction

Hong Kong operates one of the world’s most rigid exchange rate regimes: a currency board pegging the Hong Kong dollar to the US dollar at a narrow band since 1983. Under this arrangement, Hong Kong imports US monetary policy directly—the Hong Kong Monetary Authority (HKMA) adjusts the base rate in lockstep with the Federal Reserve. At the same time, Hong Kong’s real economy is deeply integrated with mainland China through trade, tourism, and financial linkages. This creates a natural tension: the interest rate channel is anchored to US conditions, while the real activity channel is driven by Chinese demand.

This paper quantifies how US monetary policy shocks and China growth shocks propagate to Hong Kong’s domestic economy. Using a six-variable quarterly VAR estimated on 1998Q1–2024Q3 data, I compute orthogonalized impulse response functions, forecast error variance decompositions (FEVD), and historical decompositions under a Cholesky identification scheme that orders external variables first.

The main findings are:

1. **China dominates real activity.** China GDP growth explains 60–78% of Hong Kong GDP forecast error variance at the 8-quarter horizon, depending on VAR vs. BVAR specification. The US federal funds rate explains less than 5%.
2. **The US dominates interest rates.** Through the currency board mechanism, the US FFR explains 59–75% of 3-month HIBOR variance, while China growth shocks contribute less than 20%.
3. **Asymmetric transmission.** US monetary tightening raises HIBOR almost one-for-one but has modest direct effects on GDP and unemployment. China slowdowns, by contrast, depress GDP and raise unemployment without an equivalent interest rate offset.

These results have direct policy relevance for HKMA’s macroprudential toolkit: since the interest rate channel is imported and the dominant real shock source (China) cannot be offset by domestic monetary policy, fiscal and prudential instruments must bear the stabilization burden.

The remainder of the paper is organized as follows. Section 2 reviews related work on small open economy VARs and Hong Kong macro modelling. Section 3 describes the data and variable construction. Section 4 presents the econometric framework. Section 5 reports the main results. Section 6 checks robustness. Section 7 concludes.

2 Literature Review

This paper contributes to three strands of literature.

Small open economy VARs. The structural VAR framework for analyzing external shock transmission to small open economies originates with Cushman and Zha (1997) and Kim (2001). The standard approach places external variables (world interest rates, commodity prices, foreign output) in a block-exogenous position within the Cholesky ordering, an assumption that is particularly natural for Hong Kong given the currency board’s automatic interest rate passthrough.

China–Hong Kong economic linkages. Genberg and He (2009) study monetary transmission under the HKD peg and find that US monetary conditions dominate short-term interest rates but have limited direct effects on output. More recently, He et al. (2014) examine how mainland economic conditions increasingly shape Hong Kong’s business cycle. My contribution is to provide a unified FEVD-based quantification of both channels simultaneously.

Currency board macro dynamics. The macroeconomics of currency boards is surveyed by Hanke (2002). Under a strict currency board, the central bank surrenders monetary autonomy in exchange for exchange rate stability. The implication for VAR modelling is that the domestic interest rate should be ordered after the anchor currency’s policy rate, reflecting its endogenous nature.

3 Data

3.1 Variables and Sources

I use six quarterly variables spanning 1998Q1–2024Q3 (107 observations):

Table 1: Variable Definitions and Sources

Variable	Description	Unit	Source
us_ffr	US effective federal funds rate	%	FRED
china_gdp	China real GDP growth	YoY %	World Bank
gdp_growth	Hong Kong real GDP growth	YoY %	World Bank
cpi_inflation	Hong Kong CPI inflation	YoY %	World Bank
unemployment	Hong Kong unemployment rate	%	World Bank
hibor_3m	3-month HIBOR	%	Derived (FFR + spread)

The US federal funds rate is obtained as a monthly series from FRED and aggregated to quarterly means. Hong Kong and China macro variables are sourced from the World

Bank’s annual indicators and interpolated to quarterly frequency using cubic spline temporal disaggregation. HIBOR is proxied as the US FFR plus a calibrated spread, reflecting the currency board’s near-automatic passthrough.

3.2 Stationarity and Cointegration

I test each variable for stationarity using both ADF and KPSS tests. Where both tests agree on a unit root, I take first differences. The dual-test protocol resolves the known size-power trade-off between ADF (conservative) and KPSS (liberal for trend-stationary processes).

Johansen cointegration testing on the I(1) subset (`china_gdp`, `cpi_inflation`, `unemployment`) detects one cointegrating relationship at the 90% level. I proceed with the differenced VAR but note that a VECM specification may improve long-horizon forecasts.

4 Econometric Framework

4.1 VAR Specification

The reduced-form VAR(p) is:

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \cdots + A_p y_{t-p} + u_t, \quad u_t \sim \mathcal{N}(0, \Sigma_u)$$

where y_t is the $K \times 1$ vector of endogenous variables ($K = 6$), c is a vector of intercepts, A_ℓ are $K \times K$ coefficient matrices, and u_t are reduced-form residuals. Lag order p is selected by BIC subject to a parameter-count guardrail ($p = 3$).

4.2 Structural Identification

I identify structural shocks via a Cholesky decomposition of $\Sigma_u = PP'$, where P is lower-triangular. The variable ordering follows the external-first convention:

[us_ffr, china_gdp, gdp_growth, cpi_inflation, unemployment, hibor_3m]

This ordering assumes that (i) US and China variables do not respond contemporaneously to Hong Kong shocks (small open economy assumption), and (ii) domestic real variables (GDP, CPI) respond within the quarter to interest rate shocks but not vice versa.

4.3 Impulse Response Functions

Orthogonalized IRFs are computed as:

$$\Theta_h = \Phi_h P, \quad \text{where } \Phi_h = JF^h J',$$

with F the companion matrix, $J = [I_K \ 0 \ \cdots \ 0]$, and $P = \text{chol}(\Sigma_u)$.

4.4 Forecast Error Variance Decomposition

The FEVD at horizon h for variable i due to shock j is:

$$\text{FEVD}_{i,j}(h) = \frac{\sum_{s=0}^h \theta_{ij,s}^2}{\sum_{s=0}^h \sum_{k=1}^K \theta_{ik,s}^2}$$

where $\theta_{ij,s}$ is the (i, j) element of Θ_s . By construction, $\sum_j \text{FEVD}_{i,j}(h) = 1$.

4.5 Historical Decomposition

I decompose each observed period into contributions from each structural shock:

$$y_t - \bar{y} = \sum_{j=1}^K \sum_{s=0}^{t-1} \Theta_s e_{j,t-s}$$

where $e_t = P^{-1}u_t$ are the structural innovations and $\bar{y}$ is the deterministic baseline.

4.6 Bayesian VAR

As a robustness check, I estimate a Minnesota-prior BVAR. The prior shrinks cross-variable coefficients toward zero and own-lag coefficients toward a random walk:

$$\text{Var}(A_{ij}^{(\ell)}) = \begin{cases} (\lambda_1/\ell^{\lambda_3})^2 & \text{if } i = j \\ (\lambda_1\lambda_2\hat{\sigma}_i/(\ell^{\lambda_3}\hat{\sigma}_j))^2 & \text{if } i \neq j \end{cases}$$

where $\hat{\sigma}_i$ is the residual standard deviation from a univariate AR for variable i , and $\lambda_1 = 0.2$, $\lambda_2 = 0.5$, $\lambda_3 = 1.0$ are hyperparameters.

5 Results

5.1 FEVD: Who Drives What?

Table 2 presents the forecast error variance decomposition at the 8-quarter horizon for both VAR and BVAR specifications.

Table 2: Forecast Error Variance Decomposition at $h = 8$ Quarters

Target	VAR (BIC, $p = 3$)		BVAR (Minnesota, $p = 3$)	
	US FFR	China GDP	US FFR	China GDP
HK GDP growth	3.0%	77.5%	4.1%	59.7%
HK CPI inflation	2.7%	28.7%	0.5%	7.8%
HK Unemployment	9.2%	26.8%	5.8%	6.7%
3M HIBOR	59.2%	18.9%	75.2%	0.9%

The central finding is a sharp division of influence:

- China GDP growth is the dominant driver of Hong Kong’s real activity (60–78% of GDP variance). The US federal funds rate contributes less than 5%.
- The pattern reverses for interest rates: the US FFR explains 59–75% of HIBOR variance, confirming the currency board’s mechanical passthrough.
- CPI and unemployment receive mixed influence from both sources, with China growth mattering more in the OLS VAR specification.

5.2 Historical Decomposition

The historical decomposition identifies which shocks drove specific episodes. During the 2008–09 Global Financial Crisis, the China growth shock (negative) accounts for the largest share of Hong Kong’s GDP decline. During 2022–23, the US monetary tightening shock dominates the rise in HIBOR, while China’s post-COVID slowdown drives the real economy weakness.

5.3 Forecast Evaluation

Expanding-window out-of-sample evaluation (minimum training window: 60 quarters, horizons: 1Q and 4Q) compares the VAR against AR(1) and random walk benchmarks. At the 1-quarter horizon, the VAR outperforms benchmarks for GDP growth and unemployment. At 4 quarters, no model consistently dominates, consistent with the difficulty of long-horizon macro forecasting.

6 Robustness

6.1 Cholesky Ordering

I re-estimate the model under three orderings: (i) the baseline external-first ordering, (ii) domestic variables first, and (iii) reversed. The qualitative result—China dominates

real activity, US dominates HIBOR—is preserved across all orderings, though the exact FEVD shares shift by 5–15 percentage points.

6.2 Sub-sample Stability

I estimate on pre-GFC (1998–2009), post-GFC (2010–2024), and full-sample-excluding-COVID sub-samples. The full sample and ex-COVID models are stable (max eigenvalue < 1). The post-GFC sub-sample shows mild instability, likely reflecting the shorter sample combined with the 2022–23 rate hiking cycle.

6.3 VAR vs. BVAR

The BVAR (Minnesota prior, $\lambda_1 = 0.2$) yields qualitatively similar FEVD patterns with tighter shrinkage. The BVAR is always stable (max eigenvalue 0.91 vs. 0.99 for OLS VAR) and shows lower coefficient norms, suggesting less sensitivity to outlier observations.

7 Conclusion

This paper provides a quantitative answer to how US and China shocks propagate to Hong Kong under the currency board. The evidence shows a clean separation: China growth dominates the real economy, while US monetary policy dominates the interest rate channel. This asymmetry has direct implications for macroprudential policy design—since the currency board precludes independent monetary responses to real shocks, fiscal and regulatory instruments must bear the stabilization burden when China’s economy slows.

Several avenues for future work remain. First, real quarterly data from official Hong Kong sources (C&SD, HKMA) would strengthen the empirical foundation. Second, a VECM specification could exploit the detected cointegrating relationship for improved long-horizon forecasts. Third, time-varying parameter VARs could capture the evolving relative importance of US and China shocks as mainland financial integration deepens.

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